Stats - Variation

Measures of Variation

Overview of Measures of Variation

- Measures of variation assess how spread out a dataset is.
  - Greater spread results in larger values for range, variance, and standard deviation.
  - Increased concentration of data yields smaller values for range, variance, and standard deviation.
- A dataset with no variability produces all measures equal to zero.
  - Example: Dataset = [5, 5, 5, 5, 5, 5, 5]
  - Range: Maximum - Minimum = 5 - 5 = 0
  - Variance and Standard Deviation = 0, aligning with no variation concept.

Characteristics of Variation Measures

- None of the measures can be negative; they are always zero or positive.
  - This is due to them representing an absolute measure of distance over which data points are spread.

Coefficient of Variation (CV)

- Abbreviated as CV, it assesses relative variation.
  - Relative variation refers to the measurement based on a comparative basis, often as a percentage.
- Formula for Coefficient of Variation:
  $CV = \left(\frac{\text{Standard Deviation}}{\text{Mean}}\right) \times 100$
- Importance: Enables comparison of datasets measured in different units.
  - For instance, comparing temperature data in Fahrenheit vs. Celsius.
- Example:
  - Stock A: Mean = 50, Standard Deviation = 5
  - Stock B: Mean = 100, Standard Deviation = 5
  - If only standard deviation is considered, they appear the same.
  - Relatively, stock A has a higher CV (10%) compared to stock B (5%), indicating stock B has lower variability relative to its mean.

Calculating Coefficient of Variation

- For Stock A:
  - $CV_A = \left(\frac{5}{50}\right) \times 100 = 10\%$
- For Stock B:
  - $CV_B = \left(\frac{5}{100}\right) \times 100 = 5\%$
  - This establishes that stock A is more variable relative to its mean than stock B.

Outliers

- Defined as extreme values that can significantly affect statistical calculations (mean, range, variance).
- Method to identify outliers: Z-score method.
- Formula for Z-score:
  $Z = \frac{X-\mu}{\sigma}$
  where:
  - X = data point
  - \mu = mean
  - \sigma = standard deviation
- Steps to identify outliers:
  1. Centering: Adjust dataset to have a mean of zero.
  2. Scaling: Normalize standard deviations across the dataset.

Example of Z-score

- Given data with a mean of 150 and a standard deviation of 50, to find the Z-score for a specific data point
  - Centering:
  - Minimum = 0, Maximum = 300
  - Shift mean by subtracting 150 from each point, resulting in:
    - For data point 620:
      $Z = \frac{620-490}{100} = 1.3$
  - Not considered an outlier since Z-score lies between -3 and 3.

Skewness and Kurtosis

- Skewness measures asymmetry of the distribution.
  - Left skewed (negatively skewed): Mean < Median.
  - Right skewed (positively skewed): Mean > Median.
- Kurtosis measures the sharpness of the distribution peak.
  - Leptokurtic: Higher, sharper peak (positive kurtosis).
  - Platykurtic: Flatter peak (negative kurtosis).
- Interpretation of skewness and kurtosis is important; not expected to compute, but to understand implications.

Quartiles and Five-Number Summary

- Quartiles split the dataset into four segments, each containing 25% of the data.
  - Q1: First quartile, separates lower 25% from upper 75%.
  - Q2: Second quartile (median), separates lower 50% from upper 50%.
  - Q3: Third quartile, separates lower 75% from upper 25%.
- Continuous measure of position in an ordered dataset:
  - Q1's position: $\frac{n+1}{4}$
  - Q2's position: $\frac{n+1}{2}$
  - Q3's position: $\frac{3(n+1)}{4}$
- Interquartile Range (IQR):
  - Measures the middle 50% of the data:
  - $IQR = Q3 - Q1$
- Five-Number Summary comprises: Minimum, Q1, Median, Q3, Maximum, providing a comprehensive view of data spread.

Relationships in Five-Number Summary

- Left skewed data: Difference between median and minimum > median and maximum.
- Symmetrical data: Differences are approximately equal.
- Right skewed data: Difference between median and minimum < median and maximum.